Logic in Computer Science

Zero-Order

Terms : $\mathcal{L}^{(0)} \rightarrow (LC, Con, Form)$

$LC \rightarrow [\neg, \supset, \wedge, \vee, \equiv, (,)]$ set of logical constants

$Con \rightarrow \emptyset$ non-logical constant $\boxed{LC \cap Con = \emptyset}$

Form → Set of formulae

- $Con \subseteq Form$ (if $A \in con$ then A Atomic)
- if $B \in Form$ then $\neg B \in Form$
- if $B \in Form$ & $C \in Form$:

  - $(B \wedge C) \in Form$
  - $(B \vee C) \in Form$
  - $(B \supset C) \in Form$
  - $(B \equiv C) \in Form$

Logic in Computer Science

More Sets

1 in each column
Functional : aRb & aRc then b=c

Function : Is functional & left total

$R^{-1} \subseteq B \times A$ if $R \subseteq A \times B$ switch

injective ⟶ functional

functional ⟶ injective

$\{0,1,2\}^2 = \{\boxed{\langle 0,0\rangle}, \langle 0,1\rangle, \langle 0,2\rangle$
$\langle 1,0\rangle, \boxed{\langle 1,1\rangle}, \langle 1,2\rangle$
$\langle 2,0\rangle, \langle 2,1\rangle, \boxed{\langle 2,2\rangle}$

left total & right total & injective & functional

Reflexive : aRa for all $a \in A$ (both items are the same)

Antireflexive : if there is no $a \in A$ such that aRa (both items are different)

Symmetric : if bRa for all aRb

Asymmetric : if it is not symmetric

Antisymmetric : if a=b for all aRb & bRa

Transitive : if aRc for all aRb & bRc

$f: A \to B \to f \subseteq A \times B$ & f is a function.

$\boxed{\{f: A \to B\} = B^A}$

B^A ← domain, range → B

so $[f: A \to B]$

Zero Order Logic

$\mathcal{L}^{(0)}$ → Zero order language of logic.

Ex. $\mathcal{L}^{(0)} = \langle LC, \{p,q\}, Form \rangle$

$p \in Form$ ∴ p is a formula

$q \in Form$ ∴ q is a formula

$\neg$ → negation / not

$\underbrace{\neg p \quad \neg q \quad \neg\neg p \quad \neg\neg\neg q}_{\text{is a formula}}$

$\wedge$ → conjunction / and

$\neg$ → negation / not

$\vee$ → disjunction / or $\quad [SF : Form \to 2^{Form} \setminus \{\emptyset\}]$

$\supset$ → implication

$\equiv$ → material equivalence

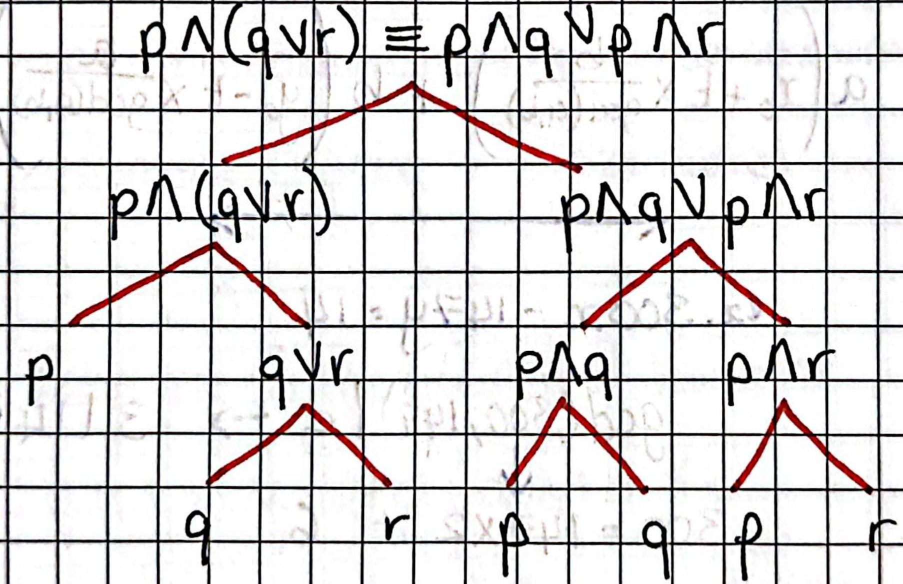
Logic in Computer Science

Zero-Order

Strongest ⟷ Weakest

$\neg \cdot \wedge \cdot \vee \cdot \supset \cdot \equiv$

Construction Tree:



$\supset$ or $\equiv$ ⟶ break from the start

$\wedge$ or $\vee$ ⟶ break from the end.

ex. $p \supset q \supset r \longrightarrow (p \supset (q \supset r))$

ex. $p \wedge q \wedge r \longrightarrow ((p \wedge q) \wedge r)$

Truth Value or Semantic Value :

Truth value of a formula → $\mathcal{F}$

with respect to given interpretation

ϱ

is denoted by

$|\mathcal{F}|_{\varrho}$

defined as

if $A \in Con$ then

$|A|_{\varrho} \rightleftarrows \varrho(A)$

if $B \in Form$ & $C \in Form$ then

$\boxed{\varrho \to \text{rho}}$

$\boxed{\Gamma \to \text{gamma}}$

- $|\neg B|_{\varrho} \rightleftarrows 1 - |B|_{\varrho}$
- $|(B \wedge C)|_{\varrho} \rightleftarrows \min(|B|_{\varrho} \cdot |C|_{\varrho})$
- $|(B \vee C)|_{\varrho} \rightleftarrows \max(|B|_{\varrho} \cdot |C|_{\varrho})$
- $|(B \supset C)|_{\varrho} \rightleftarrows \max(1 - |B|_{\varrho} \cdot |C|_{\varrho})$
- $|(B \equiv C)|_{\varrho} \rightleftarrows \begin{cases} 1 & \text{if } |B|_{\varrho} = |C|_{\varrho} \\ 0 & \text{otherwise} \end{cases}$

		$\equiv$	$\not\equiv$	$\supset$	$\vee$	$\|$	$\wedge$	$\mid$	$\bot$	$\top$
0	0	1	0	1	0	1	0	1	0	1
0	1	0	1	1	1	0	0	1	0	1
1	0	0	1	0	1	0	0	1	0	1
1	1	1	0	1	1	0	1	0	0	1

$\oplus$ → Xor

pierce's arrow

sheffer stroke

Logic in Computer Science

Valid → if it is true for all interpretations

Satisfactory → All results are true / 1

Semantic Consequence → $\Gamma \cup (\neg A)$ = unsatisfiable
($\vDash$)

Logical Equivalence → 2 formula have the same result at every interpretation.

Logic in Computer Science

Normal Form

Disjucntive Normal Form (DNF) :

- an elementary conjuction $(q \land r)$
- a disjunction of an elementary conjunction $((q \land r) \lor (r \land s) \lor (q \land s))$

Conjunctive Normal Form (CNF) :

- an elementary disjunction $(q \lor r)$
- a conjunction of an elementary disjunction $((q \lor r) \land (r \lor s) \land (q \lor s))$

properties of conjunction :

→ Commutative : $(A \land B) \Leftrightarrow (B \land A)$
→ Associative : $(A \land (B \land C)) \Leftrightarrow ((A \land B) \land C)$
→ Idempotent : $(A \land A) \Leftrightarrow A$
→ $(A \land B) \vDash A$, $(A \land B) \vDash B$
→ The law of contradiction $\vDash \neg(A \land \neg A)$ always false

$\neg(A \land B) \Leftrightarrow (\neg A \lor \neg B)$
$\neg(A \lor B) \Leftrightarrow (\neg A \land \neg B)$

always true

properties of disjunction :

→ Commutative / Associative / Idempotent (Check above)
→ $A \vDash (A \lor B)$
→ $\{(A \lor B), (\neg A)\} \vDash B$
→ Law of excluded middle $\vDash (A \lor \neg A)$ → always true

Two laws of Distributivity : $(A \lor (B \land C)) \Leftrightarrow ((A \lor B) \land (A \lor C))$
Properties of absorption : $(A \land (B \lor A)) \Leftrightarrow A$

Elementory Conjunction → If a formula is a literal or a conjunction of literals

Elementary Disjunction → If a formula is a literal or a disjunction of literals.

positive.
p & $\neg p$ → negative
Are Literals

DNF & CNF

Turning Formula into DNF & CNF:

① Draw the truth table

→ DNF ←

a	b	c	$(\neg a \supset b) \wedge (b \wedge c)$	
0	0	0	0 →	—
0	0	1	0 →	—
0	1	0	0 →	—
0	1	1	1 →	$\neg a \wedge b \wedge c$
1	0	0	0 →	—
1	0	1	0 →	—
1	1	0	0 →	—
1	1	1	1 →	$a \wedge b \wedge c$

$$\boxed{(\neg a \wedge b \wedge c) \vee (a \wedge b \wedge c)}$$

For CNF

① Construct DNF but negated
② Negate the formula again to get the original.
③ Continue using De Morgan's Laws till it's in a CNF.

$$\boxed{\begin{array}{l} \neg(A \wedge B) \Leftrightarrow (\neg A \vee \neg B) \\ \neg(A \vee B) \Leftrightarrow (\neg A \wedge \neg B) \end{array}}$$

$(A \supset B) \Leftrightarrow \neg(A \wedge \neg B) \Leftrightarrow (\neg A \vee B)$

$(A \wedge B) \Leftrightarrow \neg(A \supset \neg B)$

$(A \vee B) \Leftrightarrow (\neg A \supset B)$

$(A \vee B) \Leftrightarrow \neg(\neg A \wedge \neg B)$

$(A \wedge B) \Leftrightarrow \neg(\neg A \vee \neg B)$

$(A \equiv B) \Leftrightarrow ((A \supset B) \wedge (B \supset A))$

Note: $A \wedge (B \vee C) \Leftrightarrow (A \wedge B) \vee (A \wedge C)$

Sequents

$$\overbrace{A_1 \ldots A_n}^{\wedge} \vdash \overbrace{B_1 \ldots B_n}^{\vee}$$

hypothesis — turnstile / $\rightarrow$ can also be used — conclusion

Note: $\vdash \Leftrightarrow \supset$

$\Gamma \rightarrow$ gamma $\rightarrow$ hypothesis that aren't relevant

$\Delta \rightarrow$ delta $\rightarrow$ conclusions that aren't relevant.

Axiom $\rightarrow$ When the same atomic value appears on both sides of the turnstile. [$\Gamma, \underline{A} \vdash \underline{A}, \Delta$]

Proof Tree $\rightarrow$ A tree who's nodes are sequents & where each parent-child fragment is an instance of a proof rule.

	left	right
$\wedge$	$\dfrac{A, B \vdash}{A \wedge B \vdash}$	$\dfrac{\vdash A \quad , \quad \vdash B}{\vdash A \wedge B}$
$\vee$	$\dfrac{A \vdash \quad , \quad B \vdash}{A \vee B \vdash}$	$\dfrac{\vdash A, B}{\vdash A \vee B}$
$\supset$	$\dfrac{\vdash A \quad , \quad B \vdash}{A \supset B \vdash}$	$\dfrac{A \vdash B}{\vdash A \supset B}$
$\neg$	$\dfrac{\vdash A}{\neg A \vdash}$	$\dfrac{A \vdash}{\vdash \neg A}$
$\equiv$	$\dfrac{A, B \vdash \quad , \quad \vdash A, B}{A \equiv B \vdash}$	$\dfrac{A \vdash B \quad , \quad B \vdash A}{\vdash A \equiv B}$

First-Order Language

Extra LC $\{=, \exists, \forall\}$

Var $\rightarrow$ the countable infinite set of variables

+ the standard.

$F(0)$ = name parameters

$F(k) \quad k \geq 1$ = function parameters

$P(0)$ = propositional parameters

$P(k) \quad k \geq 1$ = predicate parameters

Logic in Computer Science

Theory Notes

1 Define the classical zero-order language:

- It's an ordered triple
- $L(0) = \langle LC; Con; Form \rangle$

2 Define the direct subformula:

- Atomic subformula have no subformula
- $\neg A$ has one subformula, A
- Direct subformula of $A \wedge B, A \equiv B, A \supset B$ are A & B.

3 Define the set of subformulae:

- $SF(A)$ is given by: $A \in SF(A)$
- If $A' \in SF(A)$ & B is a direct subformula of A' then $B \in SF(A)$

4 Define the construction tree:

- It's a finite ordered tree whose nodes are formula (formula is called A)
- Root of tree is A
- Node $\neg B$ has one child, B
- Formula $B \vee C$ or $B \equiv C$ or $B \supset C$ have 2 children, B & C
- All Leaves are atomic formula

1 Define the model of Γ set:

An interpretation ϱ is a model of the set formula Γ if $|A|_{\varrho} = 1$ for all $A \in \Gamma$

2 Define the model of a formula:

The model of Formula A is the model of singleton $\{A\}$

3 Define a satisfiable set of formula:

A set of formula is satisfiable if it has a model
for 1 formula A → If singleton $\{A\}$ is satisfiable.

4 Logical Consequence:

if the $\Gamma \cup \{\neg A\}$ then A is a logical consequence of formula Γ.

5 Valid Formula:

$\emptyset \models A$, the formula A is valid

6 Logical Equivalence:

If $A \models B$ & $B \models A$ $(A \Leftrightarrow B)$

1 Define firstorder term:

The set Term is given by the following inductive definition:

- $Var \cup F(0) \subset Term$
- If $f \in F(n)$ & $(n = 1,2,3,\ldots)$ & $t_1 \in Term$ & $t_2 \in Term$ then $f(t_1, t_2, \ldots, t_n) \in Term$.

2 Define the first order atomic formula:

- $P(0) \in Form$
- If $t_1 \in Term$ & $t_2 \in Term$ the $(t_1 = t_2) \in Fom$
- If $P \in P(n)$ & $(n = 1,2,3,\ldots)$ & $t_1 \in Term$ & $t_2 \in Term$ then $P(t_1, t_2, \ldots, t_n) \in Term$